Spillover Alpha: Sentiment Spillover Network × Post-Earnings Prediction

2026 FinHack Challenge — Case 4 | **Team 54** | UTD JSOM Finance Lab
Repository: https://github.com/Rebas9512/hackathon2026

Member	Email
Tu Le	tnl220001@utdallas.edu
Yixin Wei	yxw230015@utdallas.edu
Ernesto Aguilar	exa200020@utdallas.edu

Dashboard Preview

An interactive Streamlit dashboard brings the entire analysis to life — from Mag7 normalized returns to per-event drill-downs with model overlays.

Portfolio Overview — All Magnificent 7

Normalized return curves for all 7 companies with earnings date markers, model performance metrics, and feature importance at a glance. The right panel shows the active model's accuracy, event count, and gross return alongside a coefficient bar chart.

Single-Company Deep Dive — TSLA

> Drill into any ticker to see its price history, model prediction overlays, and per-event cumulative return chart. The "Regime Threshold" panel reveals how system connectedness drives the model's long/short decisions.

What This Project Does

We built a sentiment-driven prediction system that tests whether AI-related text signals can improve **post-earnings stock price prediction** for the Magnificent 7 (AAPL, MSFT, GOOGL, AMZN, NVDA, META, TSLA). The key innovation is a **cross-company sentiment spillover network** based on the Diebold-Yilmaz (2014) framework — capturing not just *what* people say about a company, but how sentiment *flows between* companies.

Prediction task: 5-trading-day cumulative return direction after earnings (binary: up/down)
Data: 91 earnings events across 7 companies × 13 quarters (2022 Q1 – 2025 Q1)
Methodology: Walk-forward expanding window — no lookahead, no data leakage

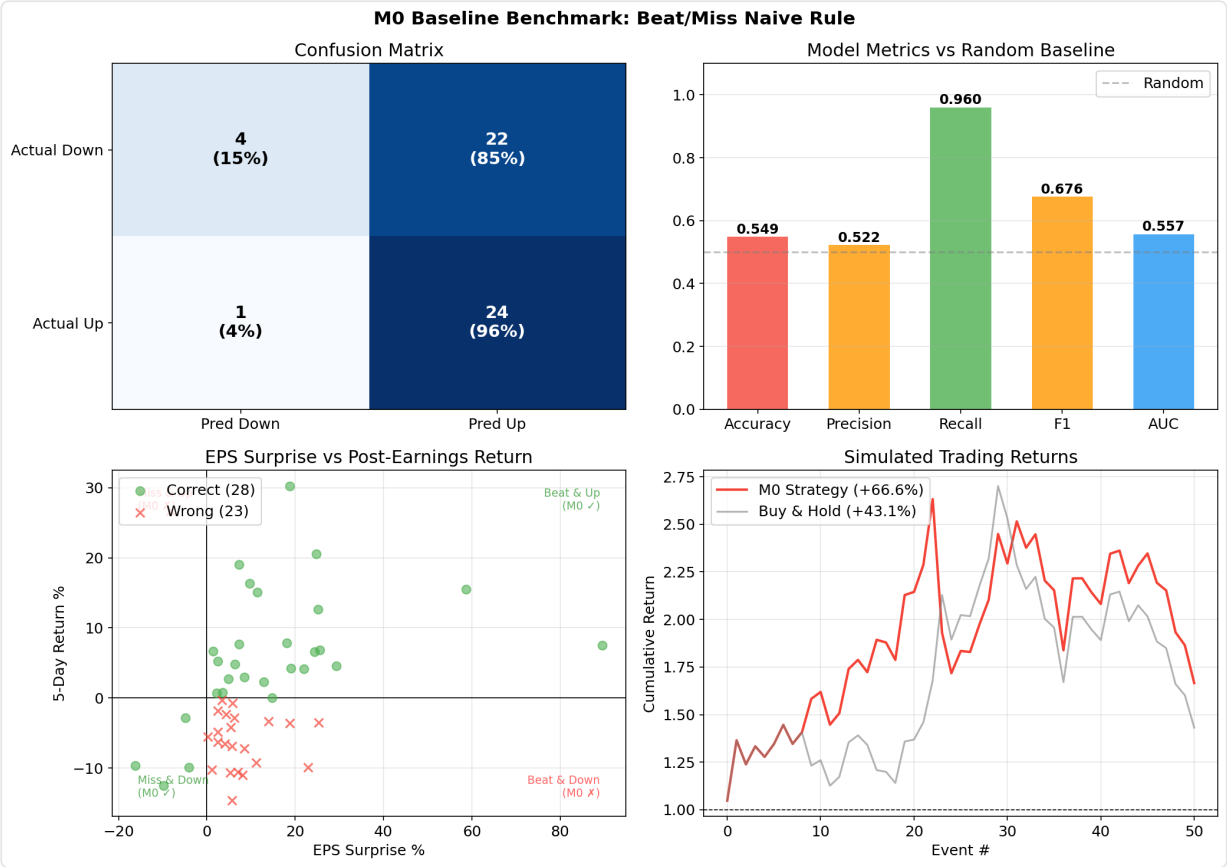
---

The Story: From Naive Rules to Network Intelligence

We tell the story through five progressive models, each adding a layer of signal.

Chapter 1 — M0: The Naive Baseline

Rule: `EPS_actual > EPS_consensus → Up, otherwise → Down`



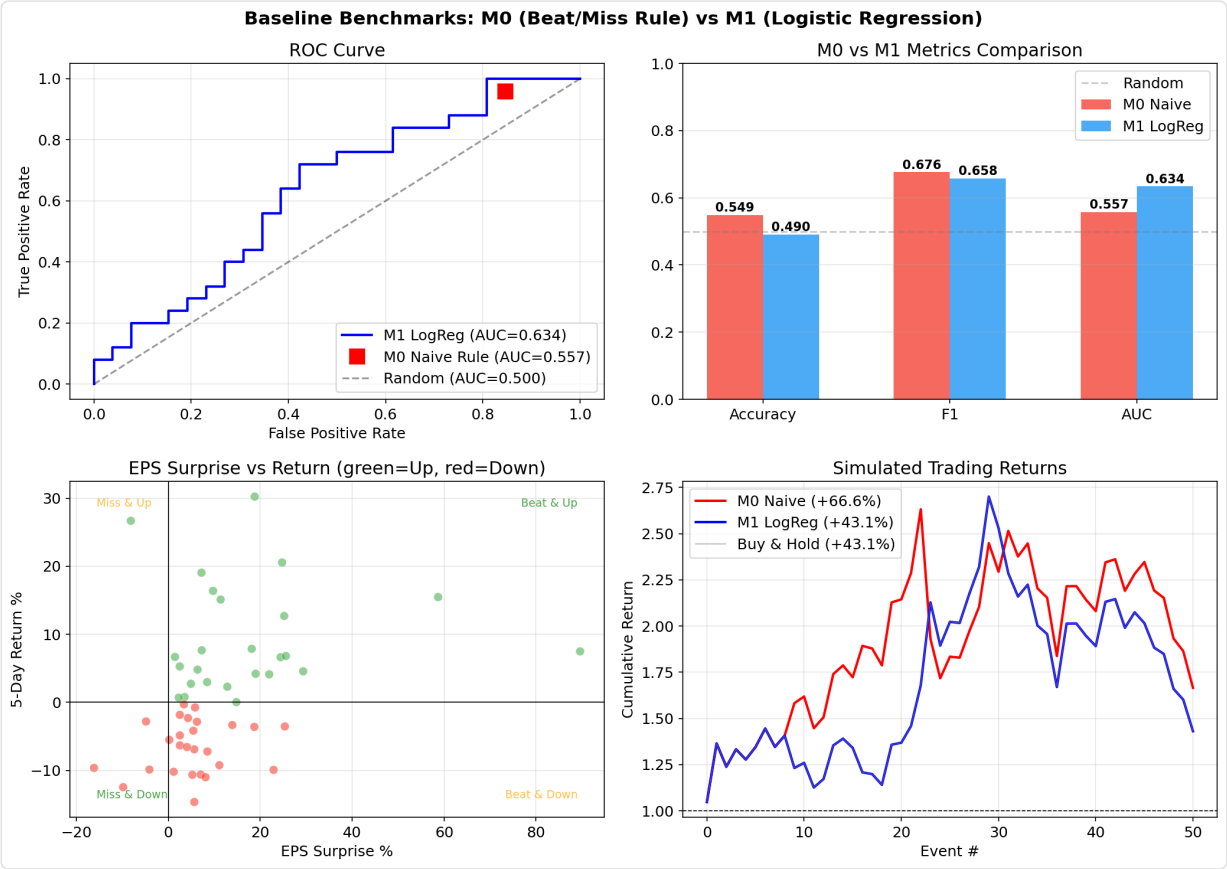
Metric	M0 (Beat/Miss Rule)	Random
Accuracy	**0.549**	0.500
AUC-ROC	0.557	0.500
Gross Return	**+66.6%**	—

The naive rule works slightly better than a coin flip (+5% over random). But near-perfect recall (0.960) with low precision (0.522) reveals a structural bias — Mag7 companies beat EPS in 90% of test events, so M0 almost always predicts "Up." Half the time, the stock drops anyway.

Why doesn't a beat guarantee an up move? The market prices in *something else* before earnings day.

Chapter 2 — M1: Does Surprise Magnitude Help?

M1 uses logistic regression on EPS surprise percentage — maybe the *size* of the beat matters.



Metric	M0 (Beat/Miss)	M1 (LogReg on Surprise %)	Random
Accuracy	**0.549**	0.490	0.500
AUC-ROC	0.557	**0.634**	0.500
Gross Return	**+66.6%**	+43.1%	—

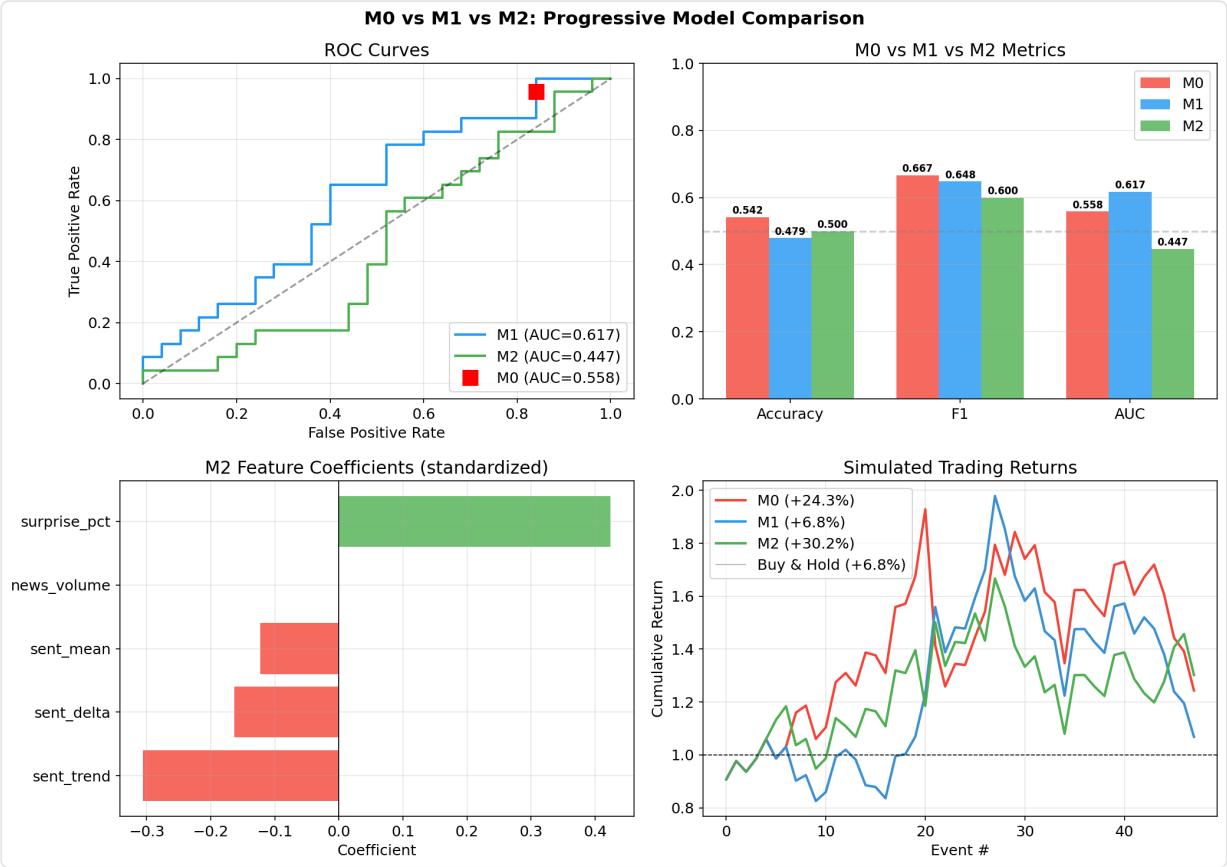
M1 has better AUC (0.634) — larger surprises *do* correlate with positive returns — but it can't find a useful decision boundary with just one feature. It predicts everything as "up", collapsing to Buy & Hold.

Takeaway: Surprise magnitude contains *ranking* information but not enough *classification* signal. The market is pricing in something beyond the earnings number itself.

Chapter 3 — M2: Adding Pre-Earnings Sentiment

Could that "something else" be sentiment? M2 adds 4 features from 7-day pre-earnings news:

- `sent_mean` : average article polarity in [ED-7, ED-1]
- `sent_trend` : sentiment slope (late vs early half)
- `sent_delta` : sentiment anomaly vs quiet period [ED-37, ED-30]
- `news_volume` : total articles in the window

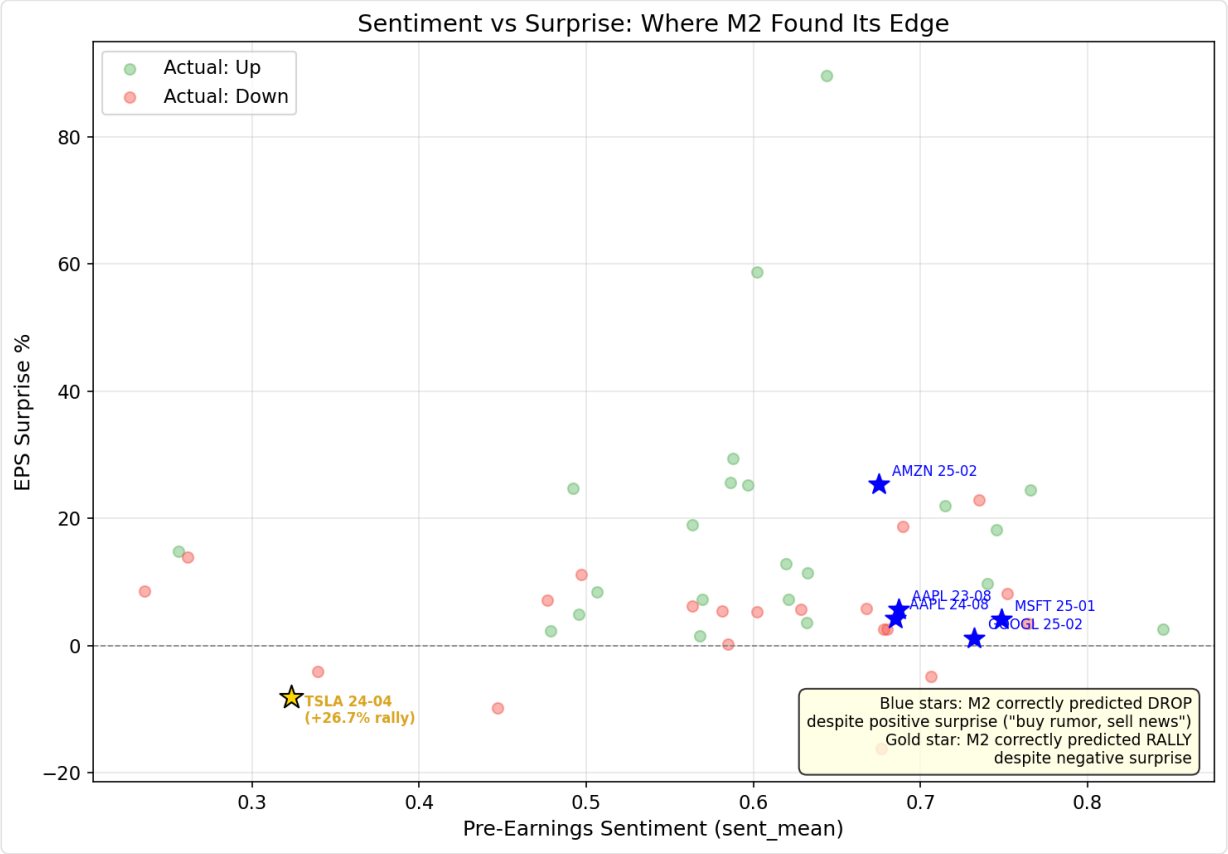


Metric	M0	M1	M2 (+ Sentiment)	Random
Accuracy	0.542	0.419	0.500	0.500
AUC-ROC	0.558	0.617	0.447	0.500
Gross Return	+24.3%	+6.8%	**+30.2%**	—

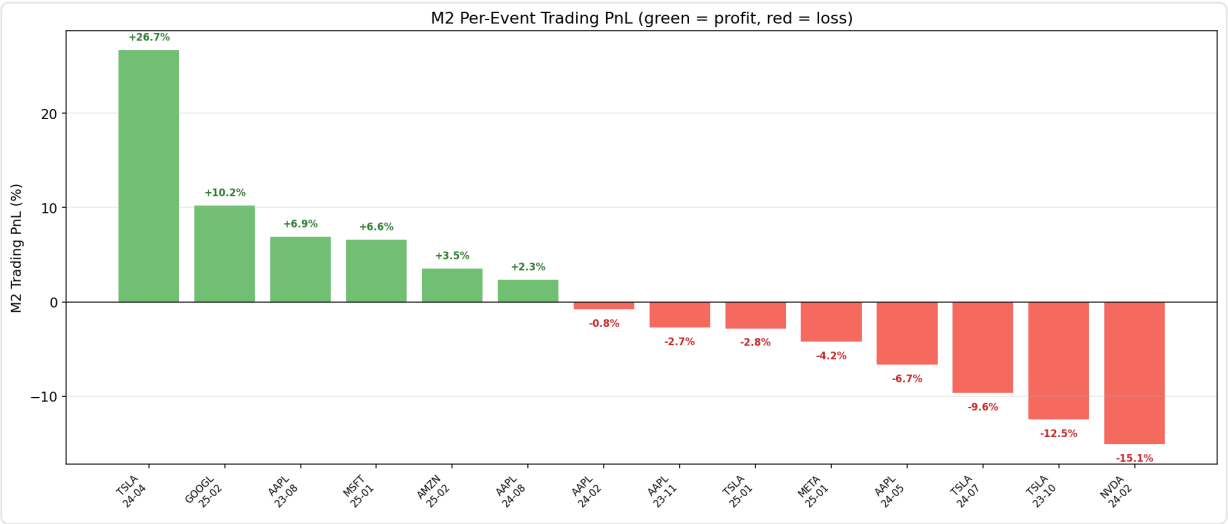
> All three models evaluated on the same 48-event test set for fair comparison.

A paradox: **M2's AUC (0.447) is worse than random**, yet it achieves the **best trading return (+30.2%)**. All 4 sentiment features have **negative coefficients** — the model learned that high pre-earnings sentiment is a *sell signal*. This is the classic **"buy the rumor, sell the news"** dynamic.

Chapter 4 — M2 Deep Dive: The "Beat But Drop" Pattern



Among 14 events where M2 and M0 disagreed, M2's profitable trades were far larger in magnitude, giving it a net **+3.6% edge**. The key victories:



5 "Beat But Drop" Victories — M2 correctly predicted DOWN despite positive EPS surprise:

Event	Surprise	5-Day Return	Sentiment	What Happened
	+5.7%	**−6.9%**	0.687	

Event	Surprise	5-Day Return	Sentiment	What Happened
AAPL 2023-08				Third consecutive quarter of declining sales overshadowed services beat
AAPL 2024-08	+4.3%	**−2.3%**	0.685	Broader tech weakness + NVDA tumble + antitrust fears
MSFT 2025-01	+4.1%	**−6.6%**	0.748	DeepSeek shock undermined AI spending thesis at peak optimism
GOOGL 2025-02	+1.2%	**−10.2%**	0.732	US-China trade war + cloud growth concerns + AI weapons controversy
AMZN 2025-02	+25.4%	**−3.5%**	0.675	Shein/Temu tariffs + cloud fears from MSFT/GOOGL misses

The Contrarian Rally — TSLA 2024-04-23: EPS surprise **−8.1%** (miss), yet stock surged **+26.7%**. Sentiment was at 0.323 (lowest in dataset). The bad news was fully priced in; Tesla's announcement of cheaper EVs triggered a massive rally. M2 saw what M0 couldn't: when sentiment is this negative, even a miss becomes a clearing event.

M2's Biggest Failure — NVDA 2024-02-21: Sentiment 0.63, M2 shorted, but NVDA was in the AI infrastructure supercycle — genuine momentum, not priced-in hype. M2 couldn't distinguish company-specific justified optimism from system-wide irrational exuberance. **This is exactly the problem M3 solves.**

Chapter 5 — M3: The Cross-Company Spillover Network

The Idea

M2's "buy the rumor, sell the news" pattern is strongest when **system-wide AI hype** is elevated. M2 fails when it can't distinguish:

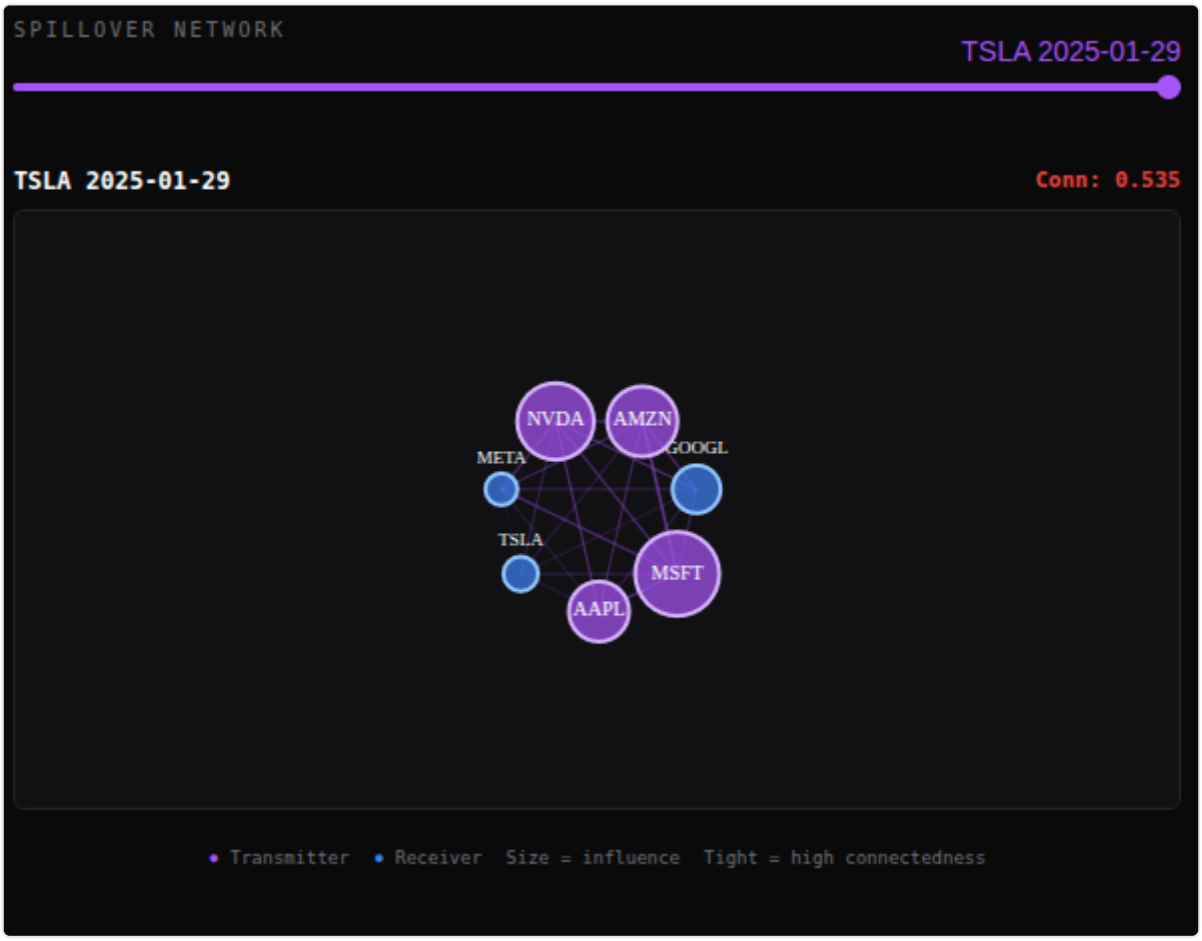
- **High sentiment in a hyped market** → priced in → sell (MSFT Jan 2025)
- **High sentiment with peers at neutral** → company-specific news → may still rally (NVDA Feb 2024)

How the Network Works

> Full technical walkthrough: see **Appendix A** below.

For each earnings event, we build a **dual-layer Diebold-Yilmaz spillover network** from ~142 days of pre-earnings data:

1. **Return layer**: VAR model on 7 Mag7 daily returns → Generalized FEVD → "what fraction of AAPL's volatility is explained by NVDA shocks?"
2. **Sentiment layer**: Same pipeline on daily sentiment → "whose sentiment shift predicts whose?"
3. **Composite**: `W = 0.5 × D_return + 0.5 × D_sentiment`



> Interactive network graph showing Mag7 spillover structure. Node size = influence strength, color = transmitter (purple) vs receiver (blue), edge thickness = spillover weight. Slider lets you explore how the network evolves across earnings events.

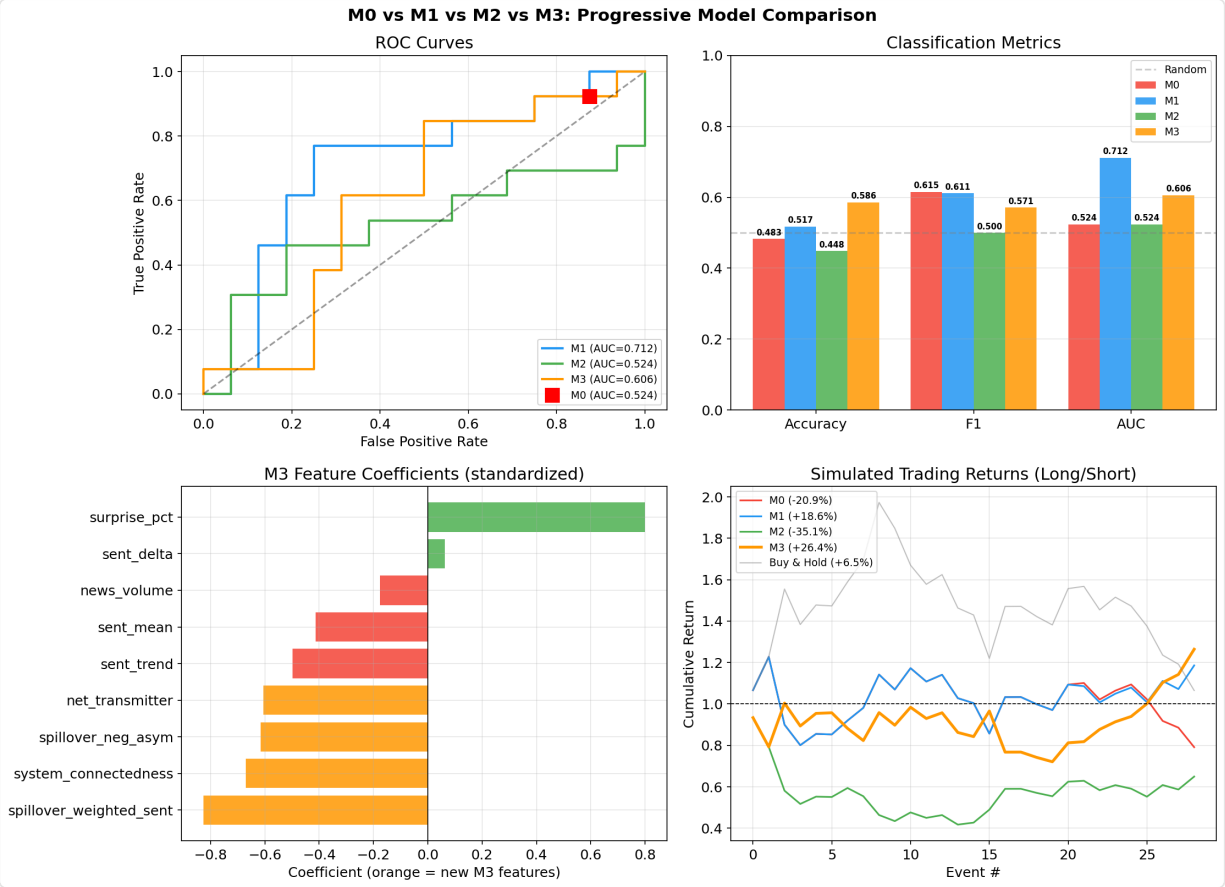
From this network we extract 4 new features:

Feature	Description
`spillover_weighted_sent`	Peers' sentiment, weighted by spillover strength into target
`net_transmitter`	Out-degree minus in-degree: leader or follower?
`system_connectedness`	How tightly coupled the Mag7 system is right now

Feature	Description
`spillover_neg_asym`	Bearish sentiment pressure from influential peers

Results: 4-Model Comparison

> All models re-evaluated on the same 29-event test set (events with complete M3 features). Numbers differ from earlier chapters due to the reduced test set — see note below.



Metric	M0	M1	M2	M3 (+ Spillover)	Random
Accuracy	0.483	0.517	0.448	**0.586**	0.500
AUC-ROC	0.524	**0.712**	0.524	0.606	0.500
Gross Return	−20.9%	+18.6%	−35.1%	**+26.4%**	—

M3 achieves the **highest accuracy (0.586)** and **best trading return (+26.4%)** — while M0 and M2 are *losing money* in this regime. The spillover features succeed precisely where M2 failed.

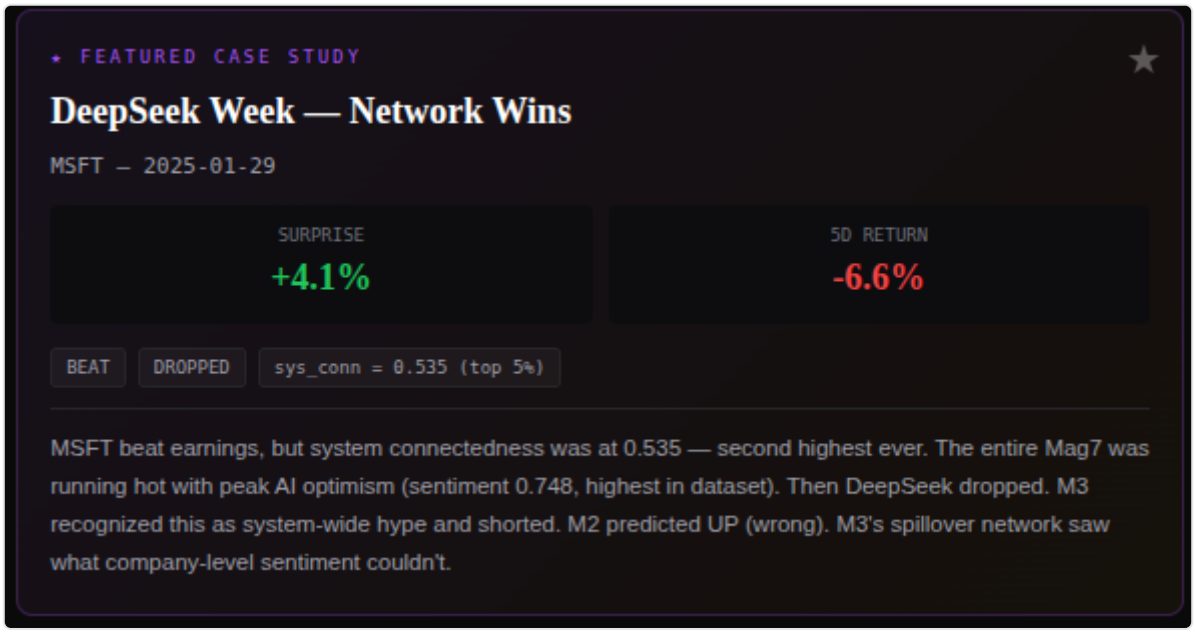
> **Why did M0 go from +66.6% to −20.9%?** The 29-event set spans late 2023–early 2025, a period of AI sentiment unwind (DeepSeek, tariffs). Mag7 companies beat EPS 90% of the

> time but stocks dropped anyway — the "beat = up" rule gets destroyed when good news is already priced in.

Chapter 6 — M3 Deep Dive: The Network Edge

Among 29 test events, M3 and M2 disagreed on **10 predictions**. M3 was correct on **7 of 10**, with a PnL edge of **+65.6%**.

Featured Case Study: DeepSeek Week

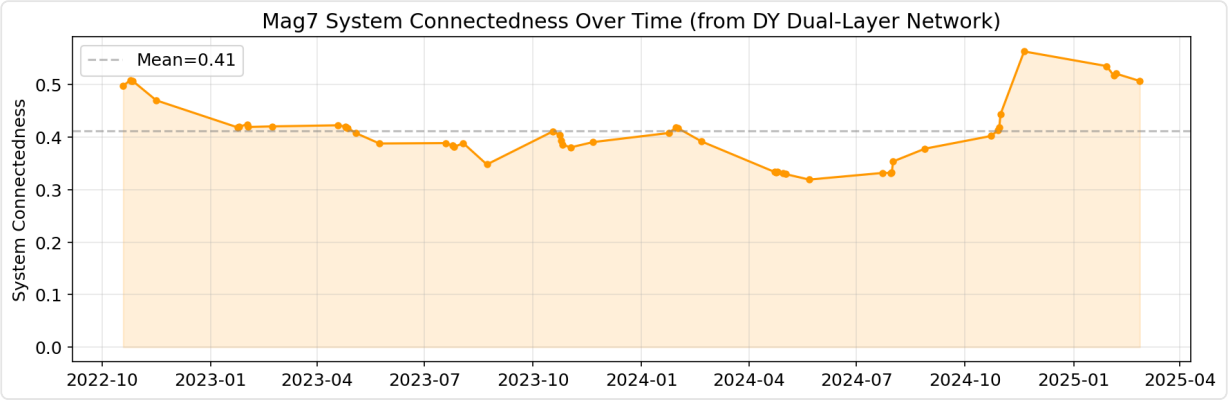


> MSFT 2025-01-29 — EPS beat +4.1%, stock dropped −6.6%. System connectedness at 0.535 (top 5%), sentiment at 0.748 (highest ever). M2 predicted UP (wrong). M3 recognized system-wide hype and shorted (correct). The DeepSeek AI breakthrough landed the same week, threatening the entire AI infrastructure spending thesis.

System Connectedness as a Regime Switch

The pattern across all M3 victories:

Regime	Connectedness	Meaning	M3's Action
High (>0.50)	Mag7 moves as one block	System-wide hype → sell the beat	Short even on EPS beat
Low (<0.35)	Mag7 is decoupled	Company-specific signal → trust it	Long if fundamentals support
Mid (0.35–0.50)	Mixed	Other features dominate	Case-by-case

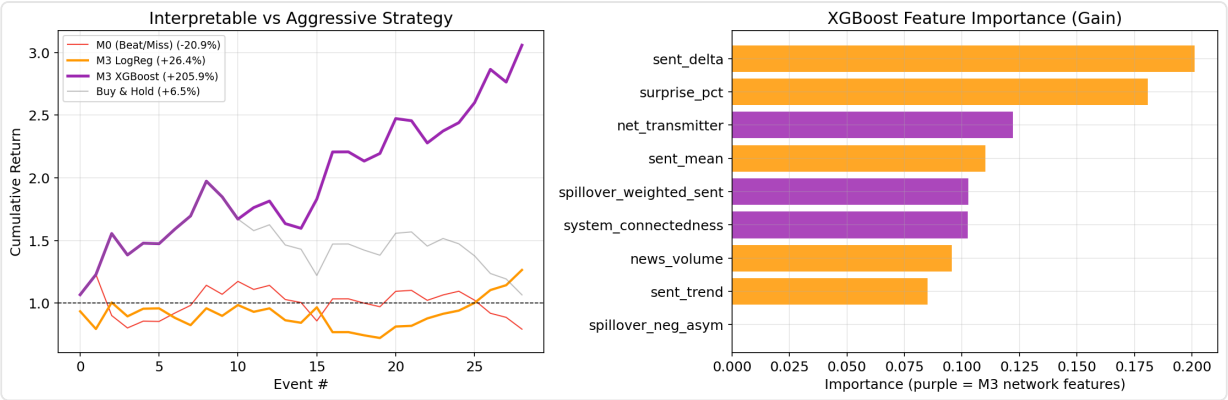


> Connectedness bottoms in mid-2024 (NVDA's idiosyncratic AI rally) and surges to peaks in Q4 2024–Q1 2025 (beat-but-drop regime).

M3's strongest feature is `spillover_weighted_sent` (coefficient −0.826) — when the companies that influence you are all highly positive, that's a sell signal. "Buy the rumor, sell the news" elevated from company-level to **network-level**.

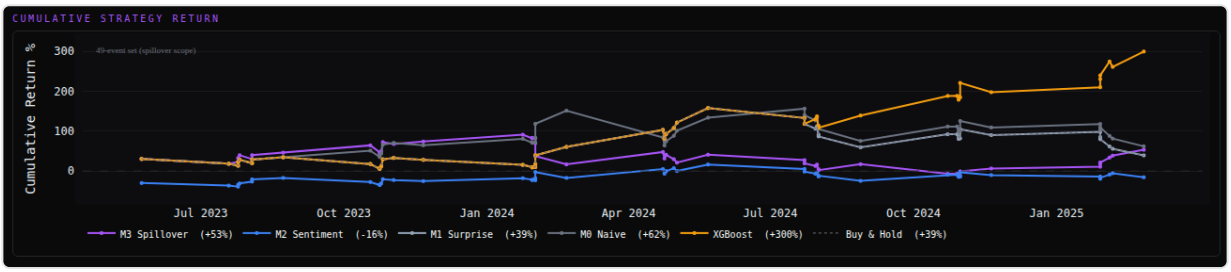
Chapter 7 — LogReg vs XGBoost: From Understanding to Trading

Logistic regression tells the *why* — every coefficient is interpretable. But the regime switch insight is inherently **non-linear**. Using the exact same 9 features, what happens with XGBoost?



Model	Accuracy	AUC	Gross Return
M3 LogReg (interpretable)	0.586	0.606	+26.4%
M3 XGBoost (aggressive)	**0.655**	**0.692**	**+205.9%**

Cumulative Strategy Returns — All Models



> Walk-forward cumulative returns for all 5 models + Buy & Hold baseline. XGBoost (orange) dramatically separates from the pack in Q4 2024 by aggressively shorting during peak system connectedness and going long in decoupled regimes.

XGBoost's edge comes from **interaction effects** logistic regression cannot represent: - "High connectedness AND high spillover sentiment → strong sell" (multiplicative) - "Low connectedness AND positive surprise → strong buy" (conditional)

On the 12 events where the two models disagreed, XGBoost was correct 7 times with a net PnL edge of +87.6%.

	XGBoost Wins	XGBoost Losses
When	Low connectedness + genuine beat	Low connectedness + genuine miss
Why	Correctly learns "decoupled + beat = buy"	Over-applies "low connectedness = always long"

> **Caveat**: With only 29 test events, the +205.9% figure carries wide confidence intervals. The interpretable LogReg model (M3) remains the primary scientific contribution — it reveals *why* spillover predicts returns. XGBoost demonstrates that exploitable non-linear structure exists, but reliable extraction requires larger samples.

Five-Model Progressive Summary

Model	Features	Purpose
M0: Naive Rule	Beat → Up, Miss → Down	"Does the surprise direction matter?"
M1: Baseline	Logistic regression on surprise %	"Does the surprise magnitude help?"
M2: + Sentiment	+ 4 pre-earnings sentiment features	"Does sentiment add predictive power?"

Model	Features	Purpose
M3: + Spillover	+ 4 dual-layer DY network signals	"Does cross-company context help?"
M3-XGB: Aggressive	Same 9 features, XGBoost (depth=2)	"Can non-linear modeling exploit the regime switch?"

Technical Details

Pipeline Design

```
Timeline per earnings event:
──[ED-150]────────[ED-37, ED-30]─────[ED-7, ED-1]──[ED]──[ED+1, ED+5]───→
     |                 |                  |          |        |
  DY Network      Quiet Period       Sentiment     Skip    Target
  Window          (baseline)         Window        Day     Variable
```

Phase	Description
Data Collection	Daily prices (yfinance), EPS (yfinance), daily sentiment (EODHD), 13,588 window articles (EODHD)
Feature Engineering	M1: surprise_pct → M2: + sent_mean, sent_trend, sent_delta, news_volume → M3: + spillover_weighted_sent, net_transmitter, system_connectedness, spillover_neg_asym
Spillover Network	Dual-layer VAR → Generalized FEVD (Pesaran-Shin 1998) on returns + sentiment, composite weight W = 0.5D_ret + 0.5D_sent
Modeling	Logistic Regression (primary) + XGBoost (robustness), walk-forward expanding window, MIN_TRAIN = 40
Evaluation	Accuracy, AUC-ROC, F1, simulated long/short trading returns

Data Leakage Controls

- Sentiment features strictly from [ED-7, ED-1]; ED-day news excluded from all features
- DY network window [ED-150, ED-8] does not overlap with sentiment window
- Walk-forward validation only — never shuffle samples
- All features use exclusively pre-ED data

Sentiment Data

Level	Source	Used For
Article-level (per-article polarity)	EODHD `/api/news`	M2 features: sent_mean, sent_trend, news_volume
Daily-aggregate (normalized sentiment)	EODHD `/api/sentiments`	DY sentiment network, quiet period baseline

---

Quick Start

```
# Install dependencies
pip install streamlit plotly pandas numpy scikit-learn xgboost networkx

# Launch the dashboard
python run_dashboard.py

# Optional: specify port
python run_dashboard.py --port 8888
```

The dashboard opens at `http://localhost:8501` with a dark terminal-style theme.

Project Structure

```
hackathon2026/
├── README.md
├── run_dashboard.py                      # Dashboard launcher
├── dashboard/
│   ├── app.py                            # Streamlit dashboard (main)
│   ├── charts.py                         # Plotly chart builders
│   └── data_loader.py                    # Data loading + walk-forward predictions
├── data/
│   ├── earnings/mag7_earnings.csv        # 91 events + EPS surprise
│   ├── prices/daily_prices.csv           # M7+SPY daily OHLCV
│   ├── sentiment/daily_sentiment.csv     # Daily aggregated sentiment
│   ├── news/window_articles.csv          # 13,588 pre-earnings articles
│   └── spillover/
│       ├── m3_features.csv               # 70 events × 4 spillover features
│       ├── connectedness_matrices.pkl    # 7×7 DY matrices
│       └── network_snapshots.json        # Network visualization data
├── notebooks/
│   ├── 00_m0_baseline.ipynb              # M0 standalone evaluation
│   ├── 01_baseline_benchmark.ipynb       # M0 vs M1 baselines
│   ├── 02_m2_sentiment.ipynb             # M2 sentiment model
│   ├── 03_m2_deep_dive.ipynb             # M2 victory/failure analysis
│   └── 04_m3_spillover.ipynb             # M3 + 4-model comparison
├── scripts/
│   ├── 01_fetch_sentiment.py
│   ├── 02_fetch_prices.py
│   ├── 03_fetch_window_news.py
│   └── 04_build_spillover_network.py     # DY dual-layer computation
├── docs/
│   └── dy_framework_explained.md         # DY framework plain-English explainer
├── outputs/                              # Generated charts and figures
├── screenshot/                           # Dashboard screenshots
├── case/                                 # Challenge case PDF
└── papers/                               # Reference papers
```

Academic References

- Diebold, F.X. and K. Yilmaz (2014). "On the Network Topology of Variance Decompositions." *Journal of Econometrics*, 182, 119-134.
- Nyakurukwa, K. and Y. Seetharam (2025). "Investor Sentiment Networks." *Financial Innovation*, 11:4.
- Pesaran, M.H. and Y. Shin (1998). "Generalized Impulse Response Analysis in Linear Multivariate Models." *Economics Letters*, 58, 17-29.

- Hutto, C.J. and E. Gilbert (2014). "VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text." *ICWSM*.

Appendix A — The Diebold-Yilmaz Spillover Network: A Technical Walkthrough

---

The Big Picture in One Sentence

We want to answer: **"When NVDA sneezes, does AAPL catch a cold?"** — and measure exactly how much of that cold came from NVDA vs. MSFT vs. TSLA vs. everyone else.

Step 0: What Our Data Looks Like

We have two spreadsheets that run every day for ~4 years (2021-06 to 2025-03):

Spreadsheet 1 — Daily Returns (from `daily_prices.csv`):

Date	AAPL	MSFT	GOOGL	AMZN	NVDA	META	TSLA
Day 1	+0.5%	+0.3%	-0.2%	+0.8%	+1.2%	+0.1%	-0.4%
Day 2	+0.2%	+0.6%	+0.4%	+0.3%	+0.8%	+0.5%	+0.1%
Day 3	-0.3%	-0.1%	-0.5%	-0.2%	-0.9%	-0.3%	-0.7%
...	...	...	...	...	...	...	...

Spreadsheet 2 — Daily Sentiment (from `daily_sentiment.csv`):

Date	AAPL	MSFT	GOOGL	AMZN	NVDA	META	TSLA
Day 1	0.74	0.81	0.62	0.55	0.90	0.43	0.32
Day 2	0.59	0.77	0.58	0.60	0.85	0.50	0.28
...	...	...	...	...	...	...	...

For each earnings event, we slice a window of ~142 days and run the full pipeline below **twice** — once on returns, once on sentiment.

Step 1: VAR — "Use Everyone's Yesterday to Predict Everyone's Today"

The Intuition

Imagine 7 kids sitting in a circle, and every day each kid picks a mood (happy or grumpy). You notice patterns:

- When NVDA was happy yesterday, AAPL tends to be happy today.
- When TSLA was grumpy yesterday, META tends to be grumpy today.
- When AAPL was happy yesterday, it's often still happy today (momentum).

A **VAR model** is just writing these patterns down as equations. For **each** company, we write:

```
Today's AAPL = (some weight × Yesterday's AAPL)
             + (some weight × Yesterday's MSFT)
             + (some weight × Yesterday's GOOGL)
             + (some weight × Yesterday's AMZN)
             + (some weight × Yesterday's NVDA)
             + (some weight × Yesterday's META)
             + (some weight × Yesterday's TSLA)
             + random noise
```

We write 7 of these equations (one per company), each with 7 weights. That's **49 weights** total for VAR(1).

What "VAR(p)" Means

The number in parentheses is how many days back we look:

- **VAR(1)**: only use yesterday → 7×7 = 49 weights
- **VAR(2)**: use yesterday AND the day before → 7×14 = 98 weights
- **VAR(3)**: three days back → 7×21 = 147 weights

More lags = more memory, but also more weights to estimate. With ~142 days of data and 7 variables, VAR(1) or VAR(2) is the sweet spot. We use BIC (a statistical score that penalizes complexity) to pick between them.

How We Estimate the Weights

Plain old linear regression (OLS), one equation at a time. Nothing fancy. With 7 variables and ~140 observations, this is a very standard, well-behaved regression problem.

The Math (For Reference)

A VAR(p) model in compact notation:

```
y_t = c + A₁·y_{t-1} + A₂·y_{t-2} + ... + Aₚ·y_{t-p} + u_t
```

Where: - `y_t` is a 7×1 vector (today's values for all 7 companies) - `A₁, A₂, ...` are 7×7 coefficient matrices (the "weights") - `u_t` is a 7×1 noise vector with covariance matrix `Σ` - `c` is a constant (intercept)

That's it. It's just multivariate linear regression.

Step 2: The Thought Experiment — "What If NVDA Gets Surprised?"

Now the VAR model is estimated. We know all the weights. Time for a thought experiment.

The Setup

Imagine it's a normal day. Suddenly, **NVDA gets a shock** — some unexpected news that moves it. Nobody else gets any direct shock. What happens next?

Because the VAR model says "today's MSFT partly depends on yesterday's NVDA", that NVDA shock starts **rippling through the system**:

```
Day 0:  NVDA gets shocked. Everyone else: nothing yet.
Day 1:  MSFT, AAPL, etc. move a little (because they partly depend on yesterday's NVDA).
Day 2:  The ripple continues — now companies react to Day 1's movements,
        which were themselves caused by NVDA's original shock.
Day 3:  Ripple gets weaker...
...
Day 10: Ripple is mostly gone.
```

This ripple is called an **Impulse Response Function (IRF)** — we "impulse" one variable and watch the "response" of all others.

What We Actually Measure: Forecast Error Variance

Instead of tracking the ripple's direction (up or down), we track its **size** (how much wobbling it causes). Specifically:

> Over the next H days, AAPL's actual values will deviate from our VAR forecast. Some of that deviation is due to AAPL's own future shocks. Some is due to NVDA's original shock still rippling through. **What fraction of AAPL's total forecast wobble came from NVDA?**

That fraction is `d_{AAPL,NVDA}`. If it's 0.15, it means: "15% of the uncertainty in predicting AAPL over the next H days can be blamed on shocks originating from NVDA."

Step 3: Why "Generalized"? — The Ordering Problem

The Old Way (Cholesky): Order Matters

The original Diebold-Yilmaz (2009) method used something called Cholesky decomposition. The problem: you have to decide an **ordering** of the 7 companies. The results change depending on whether you put AAPL first or NVDA first.

Think of it like this: if two kids sneeze at the same time and a third kid gets sick, who do you blame? Cholesky says "blame whoever I listed first." That's arbitrary.

The New Way (Pesaran-Shin 1998): Order Doesn't Matter

The Generalized approach (used in Diebold-Yilmaz 2012, 2014) does something smarter:

> For each company, pretend it's the ONLY one that got shocked. When computing NVDA's shock, allow that NVDA's shock might be slightly correlated with others (because in real life, shocks are correlated). But don't force an ordering — treat each company's shock experiment independently.

The trade-off: since we're running 7 independent experiments, the fractions for each row might not add up to exactly 100%. So we **normalize each row to sum to 1**:

```
d̃_{ij} = d_{ij} / Σⱼ d_{ij}
```

Now each row represents "what percentage of my forecast wobble came from each source?", and the percentages sum to 100%.

The Math (For Reference)

The generalized H-step forecast error variance decomposition:

```
θ_{ij}(H) = σ_{jj}⁻¹ · Σ_{h=0}^{H-1} (e_i' · Ψ_h · Σ · e_j)²
            ÷ Σ_{h=0}^{H-1} (e_i' · Ψ_h · Σ · Ψ_h' · e_i)
```

Where: - `Ψ_h` = the VAR's moving-average coefficient at horizon h (how shocks propagate h steps) - `Σ` = the covariance matrix of VAR residuals (how correlated the noise terms are) - `σ_{jj}` = variance of company j's noise (the diagonal element of Σ) - `e_i` = selection vector (all zeros except a 1 in position i)

Then normalize:

```
d̃_{ij}(H) = θ_{ij}(H) / Σ_{j=1}^{N} θ_{ij}(H)
```

Don't worry about memorizing this. The key idea is: **shock company j → measure how much company i wobbles → that's the (i,j) entry of the matrix.**

Step 4: The Connectedness Table = The Network

After running Step 3 for every pair (i, j), we get a **7×7 matrix**. This matrix IS the network.

Reading the Matrix

```
"Who is affected"     ← AAPL   MSFT   GOOGL   AMZN   NVDA   META   TSLA   | FROM others
    ↓                    shock  shock  shock   shock  shock  shock  shock   | (In-Degree)
AAPL is affected by:    [0.42]  0.12   0.10    0.08   0.15   0.06   0.07   |  0.58
MSFT is affected by:     0.10  [0.38]  0.12    0.09   0.20   0.05   0.06   |  0.62
GOOGL is affected by:    0.08   0.13  [0.40]   0.11   0.16   0.07   0.05   |  0.60
AMZN is affected by:     0.09   0.10   0.11   [0.41]  0.14   0.08   0.07   |  0.59
NVDA is affected by:     0.06   0.08   0.07    0.05  [0.55]  0.04   0.15   |  0.45
META is affected by:     0.11   0.09   0.13    0.12   0.10  [0.36]  0.09   |  0.64
TSLA is affected by:     0.07   0.05   0.04    0.06   0.18   0.08  [0.52]  |  0.48
                        ──────────────────────────────────────────────────
TO others (Out-Degree):  0.51   0.57   0.57    0.51   0.93   0.38   0.49
```

How to read this: - **Diagonal** `[0.42]`: AAPL explains 42% of its own wobble. The rest comes from others. - **Row = "imports"**: reading across AAPL's row tells you who affects AAPL and by how much. - **Column = "exports"**: reading down NVDA's column tells you how much NVDA affects everyone. - **In-Degree** (row sum minus diagonal): total influence AAPL *receives* from all others = 0.58 - **Out-Degree** (column sum minus diagonal): total influence NVDA *exerts* on all others = 0.93

Network Measures We Extract

Measure	Formula	Plain English
In-Degree of i	`Σ_{j≠i} d̃_{ij}`	"How much do others push me around?"
Out-Degree of j	`Σ_{i≠j} d̃_{ij}`	"How much do I push others around?"
Net Transmitter of i	Out-Degree − In-Degree	Positive = you're a leader. Negative = you're a follower.
System Connectedness	`Σ_{i≠j} d̃_{ij} / N`	"How tightly glued together is the whole Mag7 system right now?"

Step 5: From One Matrix to M3 Features

5a: Dual-Layer Design

We run the entire pipeline (VAR → GFEVD → 7×7 matrix) **twice** for each earnings event:

Layer	Input Data	What It Captures
Return Network	7 companies' daily stock returns	Stock price co-movement and lead-lag relationships
Sentiment Network	7 companies' daily sentiment scores	News narrative contagion — whose sentiment shifts predict whose?

Then we combine:

```
W_{ij} = 0.5 × D_return_{ij} + 0.5 × D_sentiment_{ij}
```

Why two layers? Because a company might be tightly linked to another in **price** (they move together) but loosely linked in **sentiment** (their news cycles are independent), or vice versa. The dual layer captures both channels.

5b: The Four M3 Features

For each earnings event (company `c`, date `ED`), we extract from the composite network `W`:

Feature 1: **`spillover_weighted_sent`** — "What is the mood of the companies that influence me?"

```
spillover_weighted_sent_c = Σ_{j≠c} W_{cj} × sent_mean_j
```

This takes the sentiment of every other Mag7 company and weights it by how strongly that company's shocks spill into company c. If NVDA has high spillover weight into AAPL, and NVDA's pre-earnings sentiment is very positive, then AAPL's `spillover_weighted_sent` will be pulled upward.

Feature 2: **`net_transmitter`** — "Am I leading or following the pack right now?"

```
net_transmitter_c = Out-Degree_c − In-Degree_c
```

A high positive value means this company is a **net sender** of shocks (it pushes others around more than it gets pushed). NVDA during the AI boom would score high here. A negative value means the company is a **net receiver** (it's being dragged around by the group).

Feature 3: **`system_connectedness`** — "How tightly coupled is the entire Mag7 system?"

```
system_connectedness = Σ_{i≠j} W_{ij} / 7
```

This is the same number for all 7 companies at a given point in time. When it's high, the Mag7 is behaving like one giant stock — sentiment or price shocks in any one company ripple strongly to all others. This is the "system-wide AI hype" indicator.

Why this matters for prediction: When system connectedness is high AND a company has high pre-earnings sentiment, the "buy the rumor, sell the news" effect should be strongest — because the high sentiment isn't just company-specific, it's the entire ecosystem running hot.

Feature 4: **`spillover_neg_asym`** — "Am I getting hit by negative sentiment from others?"

```
spillover_neg_asym_c = Σ_{j≠c} W_{cj} × max(0, −sent_mean_j)
```

Only counts negative sentiment from other companies, weighted by spillover strength. This captures the asymmetry that Nyakurukwa & Seetharam (2025) documented: negative sentiment is more contagious than positive sentiment. If the companies that strongly influence you are all in a bad mood, that's a stronger signal than if they're all in a good mood.

Step 6: The Timeline — What Data Goes Where

For one earnings event, say **AAPL 2024-08-01**:

```
                    ← 150 days →         ← 7d →  ED   ← 5d →
────[2024-01-15]──────────────────[2024-07-22]───[08-01]───[08-08]───→

     |_____________________________________|      |  |      |
     VAR window: [ED-150, ED-8]                  |  |      |
     ~142 trading days of returns                |  |      |
     ~142 trading days of sentiment              |  |      |
                                                 |  |      |
     Used to build the 7×7 network               |  |      Target:
                                                 |  |      5-day return
                                                 |  |      direction
                                          Sentiment |
                                          window    ED excluded
                                          [ED-7, ED-1]  (can't separate
                                          (M2 features)  pre/post news)
```

No leakage: the network is built entirely from data that was available *before* the sentiment window even begins. The network window ends at ED-8, the sentiment window starts at ED-7.

Recap: The Full Pipeline for One Earnings Event

```
1. SLICE the data
   ├── Returns: 7 tickers × 142 days → a 142×7 matrix
   └── Sentiment: 7 tickers × 142 days → a 142×7 matrix

2. FIT two VAR models (one per layer)
   ├── VAR on returns → coefficient matrices A₁ (and maybe A₂)
   └── VAR on sentiment → coefficient matrices B₁ (and maybe B₂)

3. COMPUTE two GFEVD matrices (H=10 step forecast horizon)
   ├── D_return: 7×7 matrix of "who explains whose return wobble"
   └── D_sentiment: 7×7 matrix of "who explains whose sentiment wobble"

4. COMBINE into composite network
   └── W = 0.5 × D_return + 0.5 × D_sentiment

5. EXTRACT 4 features for the target company
   ├── spillover_weighted_sent
   ├── net_transmitter
   ├── system_connectedness
   └── spillover_neg_asym

6. APPEND to the feature matrix alongside M1 and M2 features
```

Repeat for all 91 earnings events. Done.

References

- Diebold, F.X. and K. Yilmaz (2014). "On the Network Topology of Variance Decompositions." *Journal of Econometrics*, 182, 119-134.
- Diebold, F.X. and K. Yilmaz (2023). "On the Past, Present, and Future of the Diebold-Yilmaz Approach to Dynamic Network Connectedness." *Journal of Econometrics*, 50th Jubilee Issue.
- Nyakurukwa, K. and Y. Seetharam (2025). "Investor Sentiment Networks: Mapping Connectedness in DJIA Stocks." *Financial Innovation*, 11:4.
- Pesaran, M.H. and Y. Shin (1998). "Generalized Impulse Response Analysis in Linear Multivariate Models." *Economics Letters*, 58, 17-29.